Gig Economy and Changing Labor Market Dynamics

Contemporary Economic Issues: ECON 204

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Winter 2025

What is the Gig Economy?

- The **gig economy** refers to a labor market characterized by the prevalence of short-term, flexible, and freelance work rather than long-term employment.
- Workers in the gig economy typically engage in *temporary* or *part-time jobs*, often facilitated by digital platforms or mobile apps.
- Gig workers are generally considered independent contractors, not employees, which means they may not receive traditional benefits like health insurance or retirement plans.

Examples of Gig Economy Work in Canada

- **Ride-Sharing Services:** *Uber* and *Lyft* allow individuals to work as drivers on a flexible schedule.
- **Food Delivery:** Platforms such as *SkipTheDishes* and *DoorDash* enable people to deliver food as independent contractors.
- **Freelancing:** Websites like *Upwork* and *Freelancer* connect freelancers with clients seeking services in various fields such as writing, graphic design, and programming.
- **Task-Based Work:** Platforms like *TaskRabbit* offer individuals the opportunity to perform various tasks such as assembling furniture or running errands for pay.

Problem 1: Wage Disparity in the Gig Economy

A gig platform has two types of workers: high-skill and low-skill.

- High-skill workers complete 60% of tasks, earning an average of \$35/hour.
- Low-skill workers complete the remaining 40%, earning \$20/hour on average.
- Workers are paid 80% of what clients pay, and the platform retains the remaining 20%.

Questions:

1. Calculate the platform's total revenue and the wage gap between high-skill and low-skill workers if 5,000 tasks are completed in a week.
2. Assuming a 10% increase in tasks next week with the same skill distribution, what is the additional revenue earned by the platform?

Solution 1: Wage Disparity in the Gig Economy

Given Data:

- High-skill tasks: 60% of 5,000 = 3,000 tasks.
- Low-skill tasks: 40% of 5,000 = 2,000 tasks.
- High-skill wage = \$35/hour; Low-skill wage = \$20/hour.
- Workers earn 80% of client payments.

Solution 1: Wage Disparity in the Gig Economy

Solution:

$$\text{Total payment by clients} = \frac{\text{Worker earnings}}{0.8}$$
$$\text{High-skill revenue} = \frac{3,000 \times 35}{0.8} = \$131,250$$
$$\text{Low-skill revenue} = \frac{2,000 \times 20}{0.8} = \$50,000$$
$$\text{Total revenue} = \$131,250 + \$50,000 = \$181,250$$
$$\text{Wage gap} = 35 - 20 = \$15/hour.$$

10% Increase:

$$\text{New total tasks} = 5,000 \times 1.1 = 5,500$$
$$\text{Additional revenue} = 10\% \times \$181,250 = \$18,125.$$

Problem 2: Labor Supply Elasticity

In a city, the supply of gig workers follows the elasticity formula:

$$E_s = \frac{\Delta Q_s / Q_s}{\Delta W / W}$$

Given:

- Initial wage = \$20/hour; New wage = \$22/hour.
- Initial workers = 10,000; New workers = 11,500.
- Elasticity $E_s = 1.5$.

Questions:

1. Verify if the elasticity aligns with the observed data.
2. Predict the percentage increase in worker supply if wages increase to \$25/hour.

Solution 2: Labor Supply Elasticity

Verification:

$$\Delta Q_s = 11,500 - 10,000 = 1,500, \quad Q_s = 10,000$$
$$\Delta W = 22 - 20 = 2, \quad W = 20$$
$$E_s = \frac{(1,500/10,000)}{(2/20)} = \frac{0.15}{0.1} = 1.5.$$

Prediction:

$$\Delta W/W = \frac{25-20}{20} = 0.25$$
$$\Delta Q_s/Q_s = E_s \times (\Delta W/W) = 1.5 \times 0.25 = 0.375$$
$$\text{Percentage Increase} = 0.375 \times 100 = 37.5\%.$$

Problem 3: Productivity and Labor Surplus

A gig platform provides services in urban areas. The marginal productivity of each worker is given by:

$$MP = 50 - 2N$$

where N is the number of workers, and marginal productivity is measured in dollars.

- The platform pays workers a fixed wage of \$20/hour.
- Currently, 10 workers are employed.

Questions:

1. Calculate the total labor surplus (difference between marginal productivity and wages) for the platform.
2. Determine the optimal number of workers the platform should employ to maximize total labor surplus.

Solution 3: Productivity and Labor Surplus

Given Data:

- Marginal productivity: $MP = 50 - 2N$.
- Wage $= \$20/\text{hour}$.
- Current workers: $N = 10$.

Solution 3: Productivity and Labor Surplus

Labor surplus is the difference between the marginal productivity and the wage paid to the workers. For each worker, we calculate the difference between the marginal productivity and the wage, then sum it for all workers employed.

Step 1: Calculate marginal productivity for $N = 10$:

$$MP(10) = 50 - 2(10) = 50 - 20 = 30$$

Step 2: Calculate labor surplus per worker:

$$\text{Labor Surplus per Worker} = MP - W = 30 - 20 = 10$$

Solution 3: Productivity and Labor Surplus

Step 3: Calculate total labor surplus for 10 workers:

$$\text{Total Labor Surplus} = 10 \times 10 = 100$$

Thus, the total labor surplus for the platform when employing 10 workers is $\boxed{100}$.

Solution 3: Productivity and Labor Surplus

To maximize the total labor surplus, we need to maximize the total difference between the marginal productivity of labor and the wage. The total labor surplus is:

$$\text{Total Labor Surplus}(N) = N \times (MP(N) - W)$$

Substitute the expression for $MP(N)$ and W:

$$\text{Total Labor Surplus}(N) = N \times ((50 - 2N) - 20) = N \times (30 - 2N)$$

Step 1: Expand the equation:

$$\text{Total Labor Surplus}(N) = 30N - 2N^2$$

Solution 3: Productivity and Labor Surplus

Step 2: Maximize the total labor surplus by differentiating:

$$\frac{d}{dN}(30N - 2N^2) = 30 - 4N$$

Set the derivative equal to 0:

$$30 - 4N = 0$$

$$4N = 30$$

$$N = 7.5$$

Since N must be an integer, we round to the nearest whole number, so the optimal number of workers is $\boxed{8}$.

Problem 4: Gender Pay Gap in Gig Platforms

A study of a ride-hailing platform shows the following data:

- Male drivers work an average of 40 hours per week and earn \$1,200/week.
- Female drivers work an average of 30 hours per week and earn \$900/week.
- Controlling for hours worked, male drivers complete 5 rides per hour, while female drivers complete 4 rides per hour on average.
- Each ride generates \$10 in revenue.

Questions:

1. Calculate the earnings gap between male and female drivers on an hourly basis.
2. Determine how much of the gap is due to productivity differences (rides per hour) versus potential discrimination.

Solution: Gender Pay Gap in Gig Platforms

1. **Earnings Gap on an Hourly Basis:**
   - Male drivers:
   $$\text{Hourly earnings} = \frac{\$1200}{40} = \$30\,\text{per hour}$$
   - Female drivers:
   $$\text{Hourly earnings} = \frac{\$900}{30} = \$30\,\text{per hour}$$
   - **Earnings Gap:** On an hourly basis, there is no earnings gap (\$30/hour for both).

Solution: Gender Pay Gap in Gig Platforms

2. Analyzing the Gap: Productivity vs. Discrimination

- Productivity difference (rides/hour):

$$\text{Male drivers} = 5\,\text{rides/hour}, \quad \text{Female drivers} = 4\,\text{rides/hour}$$

- Revenue per ride = \$10.
- **Male's hourly revenue:**

$$5\,\text{rides/hour} \times 10 = \$50\,\text{per hour}$$

- **Female's hourly revenue:**

$$4\,\text{rides/hour} \times 10 = \$40\,\text{per hour}$$

- **Conclusion:** The productivity difference contributes \$10 per hour to the earnings gap, but no gap remains after controlling for hours worked.